

Single-Channel Fetal QRS Detection: Where the Performance Actually Comes From

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Abstract

Deep detectors for fetal QRS complexes in single-channel abdominal ECG are usually reported as a whole, so it is unclear which design choices produce the accuracy. We hold the evaluation protocol fixed (leave-one-record-out, ± 50 ms tolerance, decision threshold chosen on an inner validation record, label-blind lead selection) and vary one factor at a time. A dilated temporal convolutional network with 113 481 parameters, trained on five labour recordings of ADFECGDB, reaches a macro F1 of 99.21% on the label-blind lead and 97.45% averaged over four leads. Applied zero-shot to ten pregnancy recordings of the Silesian database (32–42 weeks, same recorder) it scores $93.30 \pm 13.46\%$; on ten records of the CinC 2013 set-a (different recorder) it falls to 59.15% (PSD lead) or 77.34% (fixed lead). A three-tier decomposition attributes the gain over a naive pipeline first to the signal front-end (10–60 Hz band: +11.00 F1), then to temporal context and per-sample output (+4.53, $p < 0.001$), and last to the architecture family at fixed context (+0.41, $p = 0.70$). Under the same protocol six architectures lie within 1.5 F1 points and none differs significantly from a dilated CNN. Against re-implemented classical baselines the network adds +11.06 points over peak picking on the same residual. A reject gate built from model confidence and rhythm regularity (cross-dataset AUROC 0.929) raises F1 by 7.4 points at 80% coverage; persistent-homology features give AUROC 0.566, a negative result we report in full.

1. Introduction

Fetal heart rate from abdominal electrodes needs a reliable fetal QRS (fQRS) detector, and a single-channel detector keeps the hardware small. The PhysioNet/CinC Challenge 2013 fixed the metric still in use, F1 with a ± 50 ms window [1], and Behar et al. benchmarked template subtraction, Kalman and blind-source methods under it, grid-searching the band-pass filter as part of the method [2]. Andreotti et al. showed with a synthetic stress-test that pre-processing choices change the ranking of extraction algorithms [3]. Since the convolutional fQRS detector of Zhong et al. [4], deep single-channel methods have been reported with many simultaneous changes: dual-path source separation trained only on synthetic data (98.08% F1 on the ADFECGDB labour group, zero-shot) [5], a classical matched-filter pipeline with power-spectral-density (PSD) lead selection ($99.5 \pm 0.5\%$ on Silesian pregnancy records, $98.0 \pm 3.0\%$ in labour, four channels) [6], and deep-learning-guided empirical mode decomposition with a 10–60 Hz band [7].

Each of these papers changes the band, the sampling rate, the temporal context, the architecture and sometimes the lead-selection rule at once, and the lead is often chosen with the test labels. None of them holds the protocol fixed and asks which of these choices carries the performance. We do that. The contributions are:

- a leakage-free single-channel protocol with a three-tier decomposition of the gain into front-end, temporal context and architecture family;
- an eight-architecture benchmark under that protocol, showing that the family does not matter while the receptive field does;
- cross-dataset evidence on 22 independent women (after a signal-level duplicate check) that generalisation breaks across recorders rather than across gestation,

plus a reject gate with a controlled negative result for topological signal-quality features.

2. Methods

2.1. Data

ADFECGDB [8]: five labour recordings (r01, r04, r07, r08, r10), four abdominal leads at 1 kHz, 5 min each, with fQRS annotations from a scalp electrode (3 191 beats). **Silesian database** [9]: ten pregnancy recordings (B1, 32–42 weeks, 20 min, 199.6 min in total) and twelve labour recordings (B2, 5 min), four abdominal leads at 500 Hz, same hospital and recorder as ADFECGDB. B1 annotations are indirect (no scalp electrode; author pipeline plus expert correction), and in 5/10 records they sit systematically 8–12 ms after the abdominal R peak; B2 annotations come from a scalp electrode. Before evaluation we cross-correlated every ADFECGDB record against every B2 record at 250 Hz over all 4×4 lead pairs and ± 5 min lags: five B2 records are the ADFECGDB records (B2_01=r01, B2_02=r10, B2_07=r04, B2_10=r07, B2_11=r08; normalised cross-correlation 0.988–0.994). The independent set is therefore 22 women (5 ADFECGDB, 10 B1, 7 new B2); the five duplicates are scored only with the fold model that never saw them. **CinC 2013 set-a** [1]: records a01–a10, four channels at 1 kHz, 1 min each, a different recorder.

2.2. Detector

One abdominal lead is band-passed 10–60 Hz (4th-order Butterworth, zero-phase) with a 50 Hz notch and resampled to 250 Hz. Maternal QRS are located on an 8–25 Hz energy envelope and removed by subtracting a median beat template with per-beat least-squares scaling. The residual and the filtered signal, each robust-scaled by median and interquartile range, form a two-channel input of 4 s (1 000 samples). The network is a dilated residual temporal convolutional network (TCN): a stem and five residual blocks

(dilations 1, 2, 4, 8, 16; 40 channels; kernel 7; BatchNorm; GELU) with a 1×1 output convolution, 113 481 trainable parameters, receptive field 379 samples (1 516 ms). It emits one logit per input sample; the target is a Gaussian heat-map ($\sigma = 3$ samples) around each annotated fQRS. Training uses binary cross-entropy with positive class weighting, AdamW (learning rate 3×10^{-3} , one-cycle schedule, weight decay 10^{-4}), six epochs, batch 32, segments cut with a 1 s stride. At inference, windows overlap by 1 s and probabilities are averaged; peaks above a threshold with a 250 ms refractory period are the fQRS. The checkpoint is 0.48 MB and one 4 s window takes 4.35 ms on a desktop CPU (920× real time).

2.3. Protocol

Leave-one-record-out (LORO) on ADFECGDB with five folds: one test record, one inner validation record (the first of the remaining four) and three training records. The decision threshold is chosen on the validation record from a grid 0.20–0.75 and applied unchanged to the test record. Detections are matched one-to-one to annotations with greedy nearest matching within ± 50 ms; macro F1 is the mean over the 20 (record×lead) evaluations. Because a clinical device sees one lead, we also report a *label-blind* lead: following Power-MF [6], the lead whose residual envelope has the largest Welch PSD peak in 1.8–3.0 Hz (108–180 bpm). This rule never touches annotations. Silesian and CinC records are scored zero-shot with the model trained on all five ADFECGDB records (threshold = median of the fold thresholds). Paired comparisons use the Wilcoxon signed-rank test on the 20 (record×lead) scores.

2.4. Experiments

Classical baselines, re-implemented and scored with the same matcher: template subtraction (TS) and TS with a two-component principal-component template (TS-PCA) [2], each followed by an adaptive Pan–Tompkins detector [10]; and *Prominence*, our own front-end followed by *scipy* peak picking (prominence above median + 3 MAD, 250 ms spacing). Hyper-parameters were tuned on r01 only and frozen for the other four records.

Three-tier decomposition. Tier 1 (front-end): eight literature band-pass settings, LORO, gradient-boosted trees on 300 ms windows of two leads. Tier 2 (temporal context and output parameterisation): a 300 ms window classifier versus the 4 s per-sample TCN, four leads. Tier 3 (architecture family at fixed context): gradient boosting versus a dilated CNN on the same 300 ms windows. A receptive-field sweep of the TCN (172–3 052 ms) isolates context within one family.

Architecture benchmark. Eight families (linear, MLP, plain CNN, dilated CNN, large CNN, CNN-GRU, TCN, ResNet-1D) trained on 300 ms windows, band 10–60 Hz, LORO with threshold chosen on the validation record, one seed, three epochs, ± 50 ms scoring.

Reject gate. Each 4 s segment is labelled *bad* if the detector’s F1 on it is below 80. A classifier is trained on 1 500 ADFECGDB segments (fold models) and tested on 600 CinC 2013 segments (zero-shot model). Feature groups: 16 topological features (Takens embedding, dimension 3, H_0/H_1 persistence statistics [11, 12], and sublevel-set H_0 of the

Table 1: Macro F1 (%) of one model, ± 50 ms, on five data configurations. SD across records. “ ≥ 90 ”/“ < 50 ”: number of records with PSD-lead F1 in that range.

Configuration	<i>n</i>	PSD lead	4-lead mean	≥ 90	< 50
ADFECDGB, LORO	5	99.21 $\pm$ 1.50	97.45 $\pm$ 2.96	5	0
Silesia B2 labour, all	12	97.17 $\pm$ 7.45	95.26 $\pm$ 9.59	11	0
7 new women, zero-shot	7	95.87 $\pm$ 9.75	93.73 $\pm$ 12.48	6	0
Silesia B1 pregnancy, zero-shot	10	93.30 $\pm$ 13.46	91.31 $\pm$ 9.99	8	0
CinC 2013 set-a, zero-shot	10	59.15 $\pm$ 37.17	61.96 $\pm$ 34.09	4	5
fixed lead 0	10	77.34 $\pm$ 28.54	–	5	3

1-D residual), 12 classical features (6 signal-only: sample entropy, kurtosis, spectral entropy, 10–60 Hz energy ratio, fetal-band PSD peak, autocorrelation lag; 6 model-output: RR coefficient of variation, plausible-RR fraction, bSQI, detection count, mean peak probability, maximum probability), and their union. The learner (logistic regression or gradient boosting) is picked by inner cross-validation on ADFECGDB only. Rejecting the least-confident segments gives a risk–coverage curve; bootstrap over 1 000 resamples.

3. Results

3.1. Detection across five data configurations

Table 1 gives the same model under the same ± 50 ms protocol on five configurations. On ADFECGDB the label-blind PSD rule reaches 99.21%, identical to the label-using oracle lead in every record, and 99.18% (Se 99.59, PPV 98.79, mean timing error 3.05 ms) on the leads prescribed by the DPSS benchmark [5]. Across four leads the LORO macro F1 is 97.43 $\pm$ 4.37% (Se 97.69, PPV 97.18, jitter 3.62 ms; 12 468 TP, 366 FP, 296 FN) and 97.16 $\pm$ 4.34% over three seeds. Re-scoring the same detections at 1 kHz gives 97.45 $\pm$ 4.44 (SD over the 20 evaluations; 2.96 over records), the value used in Tables 1 and 3.

Zero-shot on Silesian pregnancy records the model keeps 93.30%; eight of ten records exceed 90 and none falls below 50 (oracle lead 97.04%). The two weakest B1 records are those where the PSD rule picks the worst lead. On CinC 2013 the distribution is bimodal: with a fixed lead, five records are above 90 and three below 50; the PSD rule, which equalled the oracle in-domain, picks the wrong lead out-of-domain and loses 18 points (59.15 vs 77.34).

3.2. Where the performance comes from

Table 2 decomposes the gain. Changing only the band from 1–45 Hz to 10–60 Hz adds 11.00 points; the spread over eight literature bands is 18.89 points (72.17 for 0.5–100 Hz to 91.06 for 10–60 Hz). Moving from a 300 ms window classifier to the 4 s per-sample TCN adds 4.53 points ($p < 0.001$); this step bundles receptive field, output parameterisation and parameter count (33 k to 113 k). Replacing gradient boosting by a dilated CNN at the same 300 ms context adds 0.41 points ($p = 0.70$). Within the TCN family, LORO F1 rises from 92.67 $\pm$ 10.46 at a 172 ms receptive field to 97.16 $\pm$ 4.34 at 1 516 ms (three seeds, $p < 0.001$); on the validation record the sweep reads 98.73, 99.25, 99.50 and 99.52 for 172, 748, 1 516 and 3 052 ms, so the benefit saturates near 1.5 s.

Table 2: Three-tier decomposition on ADFECGDB, LORO, ± 50 ms. Tier 1 uses two leads, tiers 2–3 four leads. Paired Wilcoxon on (record $\times$ lead) scores; tier 1 was not tested pairwise.

Tier	Change	From	To	Δ F1 (p)
1 Front-end	1–45 $\rightarrow$ 10–60 Hz (GBM, 300 ms)	80.06	91.06	+11.00
2 Context	300 ms window $\rightarrow$ 4 s per-sample	92.90	97.43	+4.53 (<0.001)
3 Family	GBM $\rightarrow$ dilated CNN (300 ms)	92.49	92.90	+0.41 (0.70)

Table 3: Re-implemented baselines on ADFECGDB, ± 50 ms, hyper-parameters tuned on r01 only. Columns: macro F1 over 20 (record $\times$ lead), over the 16 evaluations excluding r01, and on the PSD lead (5 records); difference to the TCN and number of losing pairs (Wilcoxon p : TS 5.7×10^{-6} , TS-PCA and Prominence 1.9×10^{-6}). PT = adaptive Pan–Tompkins.

Method	4 leads (20)	Held-out (16)	PSD lead (5)	Δ (lost)
TS + PT	78.96 $\pm$ 23.52	75.55	87.68	−18.49 (19/20)
TS-PCA + PT	91.05 $\pm$ 10.17	90.94	96.74	−6.40 (20/20)
Prominence (our front-end)	86.39 $\pm$ 11.14	84.99	92.03	−11.06 (20/20)
FetalQRS-TCN (ours)	97.45 $\pm$ 4.44	96.81	99.21	–

3.3. Classical baselines

Table 3 shows the network gains 11.06 points over peak picking on exactly the residual it receives, winning 20/20 paired evaluations ($p = 1.9 \times 10^{-6}$); it also more than halves the spread across women and leads (SD 4.44 vs 10.17 for the best baseline). TS-PCA on the PSD lead reaches 96.74, only 2.47 points below the network, consistent with the strength of purely classical pipelines [6].

3.4. Architecture family

Figure 1 shows the eight-family benchmark. Six families (large CNN 92.43, dilated CNN 91.70, MLP 91.64, CNN-GRU 91.59, TCN 91.07, ResNet-1D 90.90) lie within 1.53 points and none differs significantly from the dilated CNN (all $p \geq 0.105$). Only the plain CNN with the same parameter count but no dilation (88.76, $p = 0.0001$) and the linear model (37.47) are different. All eight, limited to 300 ms of context, stay below the 97.43 of the 4 s per-sample model.

3.5. Reject gate and a negative topological result

Table 4 and Figure 2 give the cross-dataset reject gate. The 12 classical features predict bad segments with AUROC 0.929 and lift the retained-segment F1 from 61.96 at full coverage to 69.40 at 80% coverage (random rejection: 62.03; oracle: 74.55) and to 84.51 at 50% coverage (oracle 97.81). The strongest single features are the model’s own maximum probability (AUROC 0.970), mean peak probability (0.956), RR coefficient of variation (0.915) and the 10–60 Hz energy ratio (0.904). The 16 topological features reach AUROC 0.566, 1.4 points above random rejection at 80% coverage, and adding them to the classical set lowers AUROC to 0.905. Even the six signal-only classical features (0.666) beat topology.

Three earlier controls point the same way (3–90 Hz pilot, one validation record, threshold swept on it, so optimistic for every entry). As a detector input, a persistence-image CNN scored 27.42 where a 1-D CNN with the same parameter budget scored 97.14, and a two-stream fusion added 0.88 points ($p = 0.625$). The sublevel-set H_0 persistence of a 1-D signal equals classical peak prominence (Elder rule);

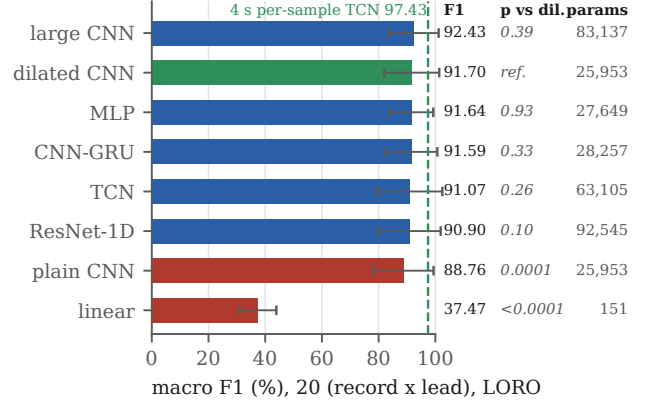


Figure 1: Eight architectures under the same protocol (10–60 Hz, 300 ms windows, LORO, threshold on validation, one seed, three epochs). Bars: macro F1 over 20 (record $\times$ lead) with SD; right-hand columns: F1, Wilcoxon p against the dilated CNN (ref.), trainable parameters. Dashed line: the 4 s per-sample TCN (97.43).

Table 4: Bad-segment prediction on CinC 2013 (600 segments) by a classifier trained on ADFECGDB (1 500 segments). F1: macro F1 of retained segments at the given coverage.

Feature set	AUROC	F1 @ 80% cov.	F1 @ 50% cov.
Random rejection	0.500	62.03	–
Topological (16)	0.566	63.40	–
Classical, signal-only (6)	0.666	–	–
Classical (12)	0.929	69.40	84.51
Combined (28)	0.905	–	–
Oracle (true F1)	–	74.55	97.81

we verified zero difference against `scipy` on 2011 peaks, so those five features are a re-parameterisation of prominence. And 8 of the 16 topological features change the sign of their Spearman correlation with segment F1 between ADFECGDB and CinC 2013 (3 of 12 classical features do). We therefore report the topological signal-quality index as a negative result.

4. Discussion

The front-end dominates. Of the 15.9 points gained over the three tiers of Table 2, 11.0 come from the pass-band alone; the same band also appears as the strongest signal-only error predictor. This matches the observation of [3] that pre-processing reorders methods, and it means that architecture comparisons run under different bands are not comparisons of architectures.

The family does not matter; the receptive field does. Six families are statistically indistinguishable at fixed context, while dilation (+2.95 at equal parameters) and a longer receptive field (172 to 1 516 ms: +4.5 LORO) are the only architectural variables with a measurable effect. Roughly one fetal RR interval and its neighbours is what the detector needs.

Generalisation breaks across recorders, not gestation. Trained on five labour records, the model loses about four points on pregnancy records from the same recorder (97 to 93) and 20–38 points on a different recorder (97 to 77 or 59). The PSD lead rule equals the oracle in-domain but col-

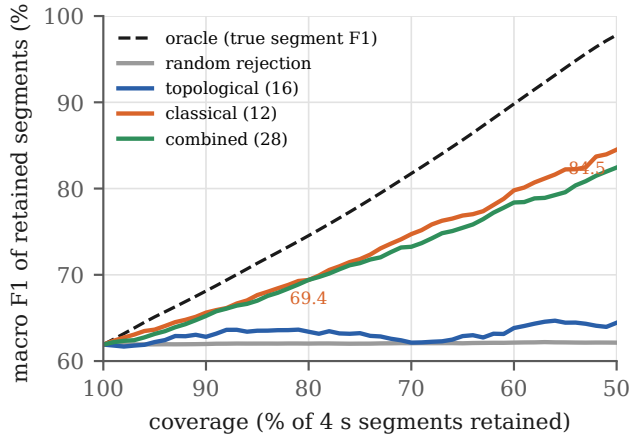


Figure 2: Risk-coverage curves on CinC 2013 for a gate trained on ADFECGDB. Dashed: oracle; grey: random rejection; blue: topological; orange: classical; green: combined. Labels: classical at 80% and 50% coverage.

lapses out-of-domain, so a learned gate should replace the spectral heuristic. The gate we tested is such a replacement: it needs no labels at test time and recovers 7.4 points at 80% coverage.

Why topology failed here. Sublevel-set persistence on a 1-D signal is prominence, which the classical set already contains; delay-embedding H_1 features depend on residual maternal amplitude and muscle noise, which differ between recorders, and their correlation sign flips; and the dominant failure mode is rhythmic (locking onto maternal beats over seconds) rather than morphological (milliseconds), so rhythm regularity and model confidence are the informative quantities. Computing cost is not the obstacle (65 ms per segment).

Limitations. Training uses five women, all in labour at 38–41 weeks, and one seed for the main model (three seeds for the receptive-field claim). B1 labels are indirect, so B1 F1 measures agreement with the authors’ pipeline, and its timing error (8.2 ms) is not comparable with B2. CinC 2013 results rest on ten 1-min records and are bimodal, so record-level confidence intervals are wide. The architecture benchmark uses one seed and three epochs. Published numbers of DPSS and Power-MF are quoted, not re-run: Power-MF needs four channels and MATLAB dependencies not in its repository. Only one topological configuration was tried, deliberately without tuning on test labels.

5. Conclusion

With protocol held fixed, single-channel fQRS detection on ADFECGDB reaches 99.21% F1 on a label-blind lead and 93.30% zero-shot on pregnancy recordings, and the gain over a naive pipeline comes from the 10–60 Hz front-end and from 1.5 s of temporal context, not from the network family. Cross-recorder transfer, not gestational age, is the open problem; a confidence-and-rhythm reject gate mitigates it, and persistent-homology features do not. Code, checkpoints and logs are available from the authors.

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